

Powered by GPT - Software Quality Verification

Final Report for Software Testing Course

System Under Test	Online Sales Management System
Repository	https://github.com/Alucard30Dec/Online-Sales-Management-System
Appendix Link	https://github.com/Alucard30Dec/Online-Sales-Management-System/tree/main/Report%20Test%20subject/SV00123-ATU-A01
Submission Date	April 11, 2026

Team Members

Name	Class	Student ID
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Record of Changes

Version	Date	Author	Change Summary
0.1	2026-04-05	QA team	Completed project audit, submission structure, and test plan baseline
0.5	2026-04-05	QA team	Added UI test cases, API test cases, test data, and automation design
0.8	2026-04-05	QA team	Added automation implementation, execution evidence, defect workflow, and metrics
1.0	2026-04-06	Hoang Van Thien	Consolidated final report content, analysis, and appendix linkage
1.1	2026-04-10	QA team	Expanded verified UI execution evidence, refreshed metrics, and synchronized the report with the updated test case baseline
1.2	2026-04-11	QA team	Removed UI Not Run status through real execution, refreshed defect log, metrics, report, and slides

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I. Overview

1.1 Project Information

This report presents the final software-testing submission for the **Online Sales Management System**, an ASP.NET Core MVC (.NET 8) application that manages authentication, products, suppliers, customers, purchases, invoices, stock, reports, and a public product catalog. The project also exposes a small API surface for service health and public catalog access. The test submission was built directly from source-code inspection, seeded demo data, local execution, and evidence collected from the running application.

The main objective of this report is to verify whether the project behaves correctly for its business-critical workflows and whether the resulting artifacts satisfy the course rubric in test planning, test design, execution evidence, reporting quality, and automation readiness.

1.2 System Under Test

- Project name: Online Sales Management System
- Technology stack: ASP.NET Core MVC (.NET 8), EF Core, ASP.NET Identity, TiDB/MySQL
- Test surfaces:
 - Admin UI
 - Public Product Catalog
 - GET /api/v1/health
 - GET /api/v1/catalog/*

1.3 Project Team And Task Allocation

The project was completed by a three-member QA team. Work allocation was designed so each member owned more than ten UI test cases and covered separate functional areas to avoid unnecessary overlap. Hoang Van Thien handled a slightly larger portion of the workload and also led the integration of planning, automation direction, and final report consolidation.

Team allocation

- Hoang Van Thien
 - led project audit and final integration
 - owned 17 UI test cases
 - main modules: Authentication, Permissions, Admin Users, Admin Groups, Invoices, Reports, partial Public Catalog
- Nguyen Thanh Dat
 - owned 13 UI test cases
 - main modules: Customers, Suppliers, Purchases, partial Stock
- Le Quang Duy
 - owned 14 UI test cases
 - main modules: Products, Product Import, partial Stock, partial Public Catalog

This allocation satisfies the requirement that each member contributes at least ten test cases while keeping role ownership visible for defense and review.

Member	Primary ownership	UI cases owned
Hoang Van Thien	Authentication, permissions, admin, invoices, reports	17
Nguyen Thanh Dat	Customers, suppliers, purchases, partial stock	13
Le Quang Duy	Products, product import, partial stock, public catalog	14

II. Test Plan

2.1 Test Scope

The testing scope focused on the highest-risk business and control areas of the project:

- authentication and login validation
- role-based access control for **admin**, **sales**, and **warehouse**
- product and master-data management
- purchase creation, receiving, and cancellation
- invoice creation, payment flow, and cancellation
- stock visibility and movement history
- report filtering and summary totals
- public catalog browsing, filtering, sorting, and details
- health and catalog API behavior

The following items remained outside scope for this submission:

- performance and load testing
- deep security penetration testing
- mobile testing
- full responsive certification across multiple devices
- unrevealed write APIs outside the exposed public catalog and health endpoints

2.2 Test Strategy And Approach

The project used a mixed strategy of **manual testing**, **API testing**, and **targeted automation**. Manual testing was prioritized for broad business-rule coverage, permission behavior, validation feedback, and human verification of visible outcomes. API testing was used for stable request-response validation on the exposed endpoints. Automation was applied selectively to high-value flows to maximize bonus potential without overextending the scope into unstable or low-value areas.

Execution followed a **black-box** approach, while test design used **white-box-informed** analysis of the source code to identify hidden edge cases, validation rules, route behavior, and permission boundaries.

2.3 Test Environment

The verified execution environment was:

- OS: Windows 11
- Runtime: .NET 8
- Browser verified: Google Chrome
- Local base URL: `http://localhost:5068`
- Repository path: `E:\Project\Online-Sales-Management-System`
- Database environment tag observed in UI: `TiDB / test`

The test data environment relied on seeded demo data from `Data/DbSeeder.cs`, including valid accounts, products, suppliers, customers, purchases, invoices, expenses, and stock movement records.

2.4 Tools Used

- planning and traceability: Markdown + Excel-compatible CSV/XLSX artifacts
- UI testing: manual browser testing + Selenium WebDriver with .NET 8 and xUnit
- API testing: Postman and Newman
- bug management preparation: GitHub Issues taxonomy and defect register
- evidence and metrics: screenshots, TRX runner output, Newman output, CSV/XLSX metrics packs

Item	Value
Operating system	Windows 11
Runtime	.NET 8
Primary browser	Google Chrome
Base URL	<code>http://localhost:5068</code>
Database tag	<code>TiDB / test</code>
Repository path	<code>E:\Project\Online-Sales-Management-System</code>

III. Test Design And Execution

3.1 Test Scenario Design

The scenario design phase identified 42 documented scenarios across UI and API surfaces. These scenarios were grouped by module and risk type, including positive flows, negative flows, validation flows, permission checks, and business-rule edge cases. The design intentionally emphasized critical modules such as authentication, permissions, purchases, invoices, stock, reports, and product import.

At the current stage, all 42 documented scenarios are represented in the current test-case files, which gives a scenario design coverage of 100%. This closes the earlier traceability gap and strengthens the design section against rubric checks for completeness and coverage.

Scenario metric	Value
Documented scenarios	42
Scenarios mapped to test cases	42
Scenario design coverage	100%
Scenarios with execution evidence	42
Scenario execution coverage	100%

3.2 UI Test Case Design

The UI test suite contains 44 test cases covering:

- authentication
- permissions
- admin users and admin groups
- customers
- suppliers
- products
- product import
- purchases
- invoices
- stock

- reports
- public catalog

Each UI test case includes:

- Test Case ID
- Scenario ID
- Title
- Module
- Preconditions
- Test Data
- Steps
- Expected Result
- Actual Result
- Status
- Owner
- Evidence / Note

The test cases were written to maximize clarity, reproducibility, and rubric coverage.

Negative cases were prioritized for authentication, permissions, validation, duplicate inputs, import restrictions, and inventory-related workflows. In the current synchronized package, every UI case now has runtime evidence and no UI row remains in Not Run.

Test Case ID	Module	Preconditions	Expected Result	Actual Result	Status
TC-UI-AUTH-001	Authentication	App is running; seeded admin account exists	Admin user reaches dashboard after valid login	Focused rerun reached the admin dashboard successfully	Pass
TC-UI-PUR-001	Purchases	Warehouse or admin is logged in; valid supplier	Draft purchase is created and redirects to	Draft purchase was created successfully	Pass

		and product exist	details page	and redirected to details page	
TC-UI-INV-001	Invoices	Sales or admin is logged in; in-stock product exists	Invoice is created successfully and stock is reduced	Create action failed and returned to the form with error toast; server log confirmed backend exception	Fail

3.3 API Test Case Design

The API test suite contains 19 cases mapped to the real endpoints found in the source code:

- GET /api/v1/health
- GET /api/v1/catalog/products
- GET /api/v1/catalog/products/{id}
- GET /api/v1/catalog/trending
- GET /api/v1/catalog/filters

The API cases cover:

- smoke validation
- happy-path catalog queries
- validation and negative inputs
- boundary values
- detail retrieval
- not-found behavior
- lookup endpoints

The API workbook also includes an **Evidence / Note** column so each API case can point back to the Newman text output and XML runner artifact used for execution proof.

All API cases were derived from real routes only. No undocumented endpoint was added.

Test Case ID	Endpoint	Expected Status	Expected Result	Actual Result	Status
TC-API-HLT-001	GET /api/v1/health	200	Health payload contains status, service, serverTime Utc, version	Newman full run passed all health assertions	Pass
TC-API-CAT-004	GET /api/v1/catalog/products?brandId=30039&page=1&pageSize=5	200	Returned items match brandId=30039 or empty array without error	Request passed all Newman assertions in full run	Pass
TC-API-CAT-012	GET /api/v1/catalog/products?sort=popularity&page=1&pageSize=5	400	Validation error indicates unsupported sort option	Request returned expected validation status and passed Newman assertions	Pass

3.4 Test Data

The project uses seeded credentials and curated datasets to avoid generic placeholder data.

Real reusable test data includes:

- valid seeded accounts for `admin`, `sales`, and `warehouse`
- invalid credential combinations
- category, brand, and product identifiers observed from the running system
- API query variations for positive and negative requests
- product import workbooks with mixed-validation rows
- invalid upload samples for attachment and file-format testing

Sensitive live credentials were not committed. Only demo-safe seeded data and controlled testing datasets were included in the repository.

Data group	Real source	Purpose
Seeded admin accounts	<code>admin@osms.local</code> , <code>sales@osms.local</code> , <code>warehouse@osms.local</code>	Auth, permission, purchase, invoice, report, and stock flows
Negative credentials	fake password / unknown email / inactive account placeholder	Invalid login and inactive-account scenarios
UI business data	prepared supplier, product, import, and form values	CRUD, purchase, invoice, and import execution
API query data	real category, brand, product IDs and negative query variations	Health and catalog API validation coverage

3.5 Automation Design And Implementation

To improve rubric score and bonus potential, a limited automation scope was implemented with maintainability in mind:

- UI automation stack: `.NET 8` + `xUnit` + `Selenium WebDriver`
- API automation stack: `Postman` + `Newman`
- design pattern: `Page Object Model`
- reusable support layer for:

- configuration
- CSV test-data loading
- waits
- screenshots
- webdriver management

The selected UI automation flows were:

- admin login smoke
- sales access-denied navigation
- privileged-user draft purchase creation
- privileged-user invoice creation
- product import preview

The selected API automation groups were:

- health smoke
- catalog happy path
- catalog validation and negative cases
- product detail
- trending and filters

Automation surface	Implemented items	Evidence type
UI Selenium + xUnit	login smoke, permission denial, purchase draft creation, invoice creation, import preview	screenshots + TRX
API Postman + Newman	health, catalog list/detail, validation, trending, filters	Newman text output + XML
Reusable design	page objects, waits, screenshot helper, CSV loader, WebDriver factory	source code in TestScript-Data/Automation/

3.6 Execution Summary

As of 2026-04-11, the synchronized execution baseline now covers all designed UI and API cases with real runtime evidence:

- total test cases: 63
- executed: 63
- pass: 56
- fail: 7
- blocked: 0
- not run: 0
- execution progress: 100%

The strongest confirmed pass results now include:

- TC-UI-AUTH-001, TC-UI-AUTH-002, TC-UI-AUTH-004, TC-UI-AUTH-005, and TC-UI-AUTH-006
- TC-UI-ADM-001 and TC-UI-ADM-002
- TC-UI-CUS-001, TC-UI-CUS-002, and TC-UI-CUS-003
- TC-UI-SUP-001
- TC-UI-PRD-001, TC-UI-PRD-002, TC-UI-PRD-003, TC-UI-PRD-004, TC-UI-PRD-005, and TC-UI-PRD-006
- TC-UI-IMP-001, TC-UI-IMP-002, and TC-UI-IMP-004
- TC-UI-PUR-001, TC-UI-PUR-004, TC-UI-PUR-005, TC-UI-PUR-006, and TC-UI-PUR-007
- TC-UI-INV-002 and TC-UI-INV-004
- TC-UI-STK-001, TC-UI-STK-002, and TC-UI-STK-003
- TC-UI-REP-001 and TC-UI-REP-002
- TC-UI-PUB-001, TC-UI-PUB-002, and TC-UI-PUB-003
- all 19 API cases - full Newman collection pass

The current confirmed fail results are:

- TC-UI-INV-001, TC-UI-INV-003, and TC-UI-INV-006
 - linked root defect: BUG-20260406-001
 - invoice creation fails before reaching the intended business validation because of the known transaction defect
- TC-UI-INV-005
 - linked defect: BUG-20260411-004
 - invoice cancellation fails and stock is not returned
- TC-UI-IMP-003
 - linked defect: BUG-20260411-003
 - valid preview confirmation still fails during import commit
- TC-UI-PUR-002 and TC-UI-PUR-003
 - linked defect: BUG-20260411-002
 - purchase validation banner renders without readable text for two required-input scenarios

Final result summary	Value
Total test cases	63
Executed	63
Pass	56
Fail	7
Blocked	0
Not Run	0
Execution progress	100%

Additional metrics	Value
Pass rate on executed cases	88.89%
Fail rate on executed cases	11.11%
Scenario count	42
Mapped scenarios	42
Executed scenarios	42
Scenario execution coverage	100%
Confirmed defects	4

3.7 Execution Evidence

The current evidence set proves that both UI and API automation are functioning beyond a smoke-only baseline and now supports full case-level execution traceability:

- UI evidence
 - login success screenshot exists for TC-UI-AUTH-001
 - permission-denial behavior screenshot exists for TC-UI-AUTH-003
 - import preview-count screenshot exists for TC-UI-IMP-002
 - purchase details screenshot exists for TC-UI-PUR-001 and TC-UI-PUR-007
 - invoice failure screenshot exists for TC-UI-INV-001
 - extended execution screenshots now exist for admin users, customers, reports, stock, public catalog, and additional invoice, purchase, and product-import scenarios
- API evidence
 - full Newman collection output exists for all 19 API requests
- server-log evidence
 - extracted defect log exists for BUG-20260406-001

Figure E-1. Admin login success evidence.

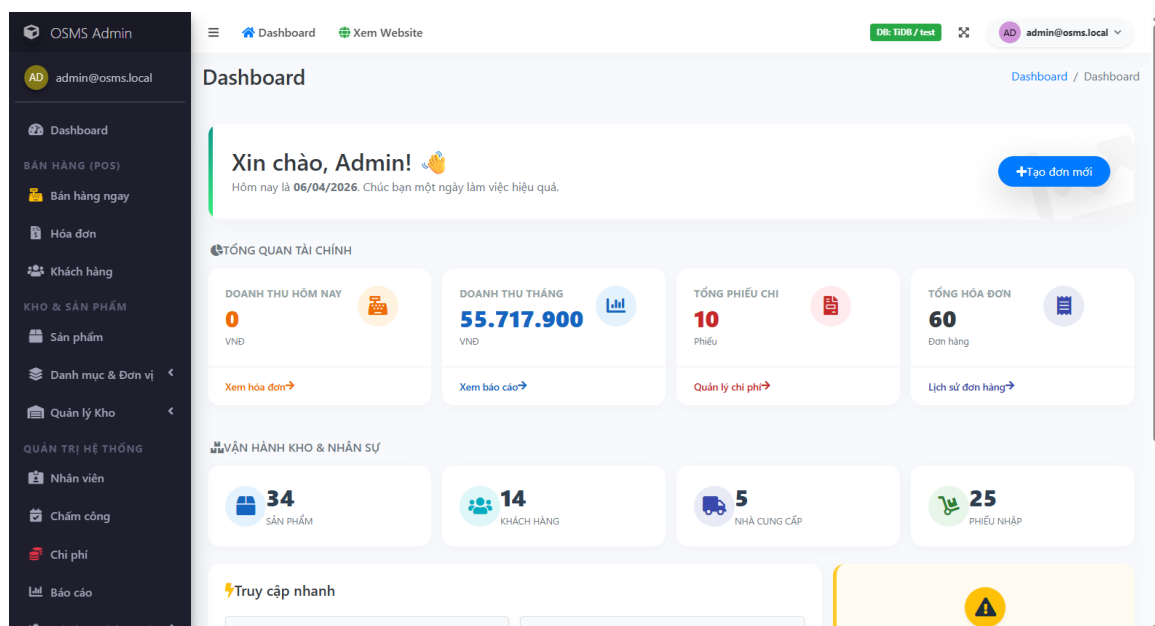


Figure E-1 - Admin login success

Figure E-2. Newman full-run summary snippet.

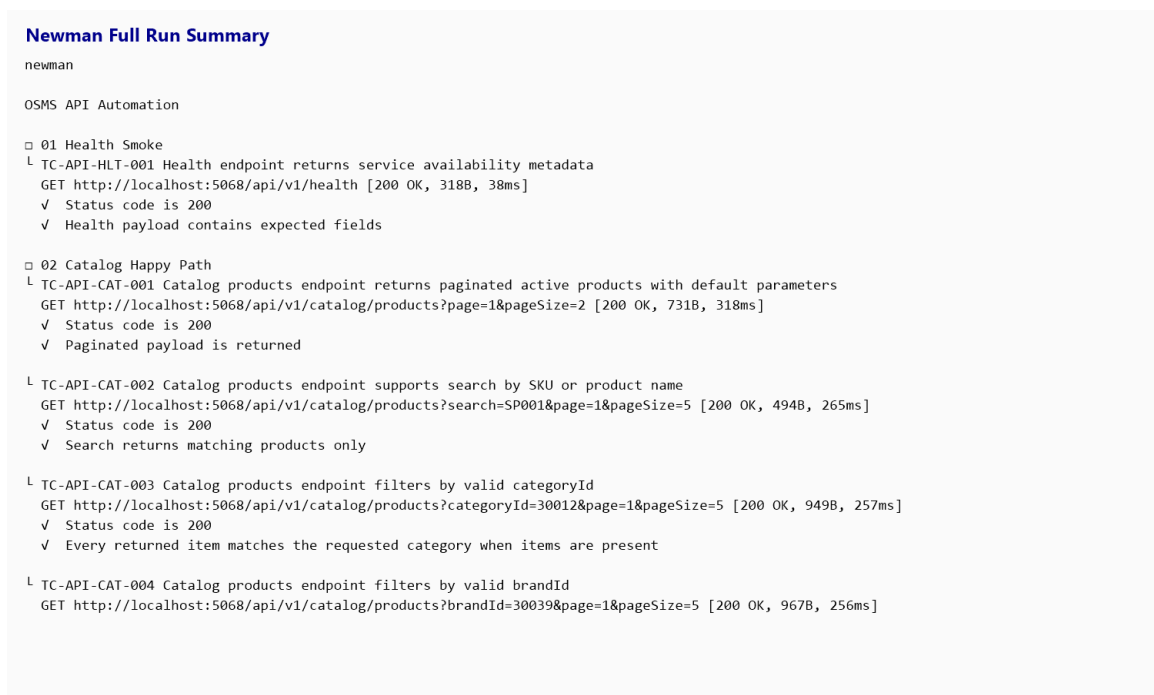


Figure E-2 - Newman full run summary

Figure E-3. Draft purchase creation success evidence.

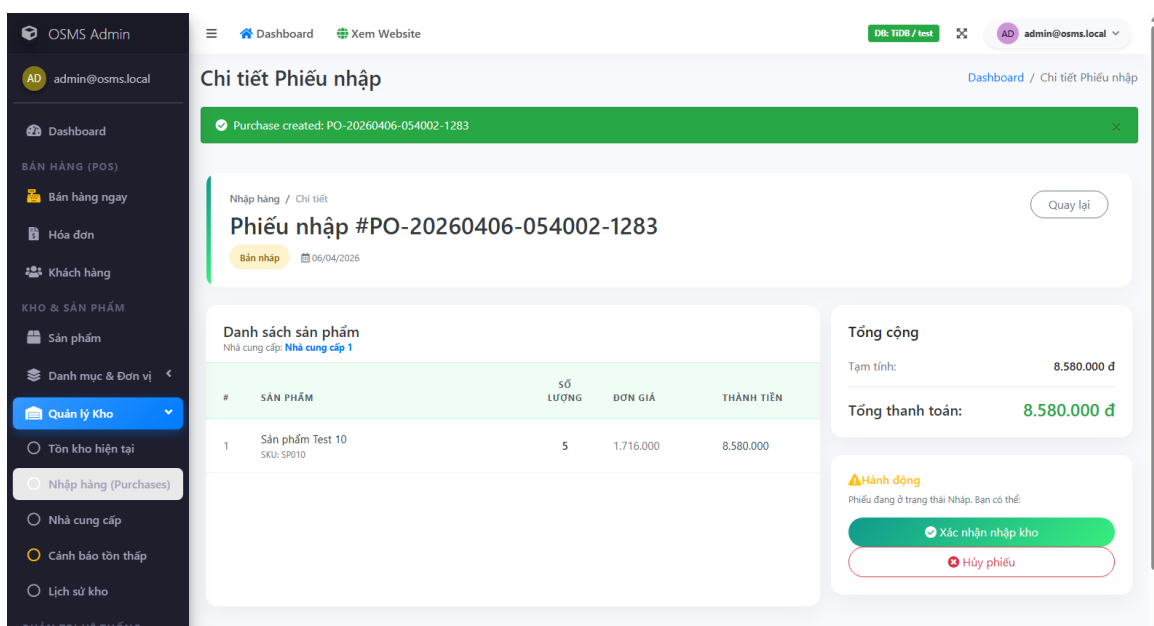


Figure E-3 - Draft purchase created

Figure E-4. Invoice creation failure evidence.

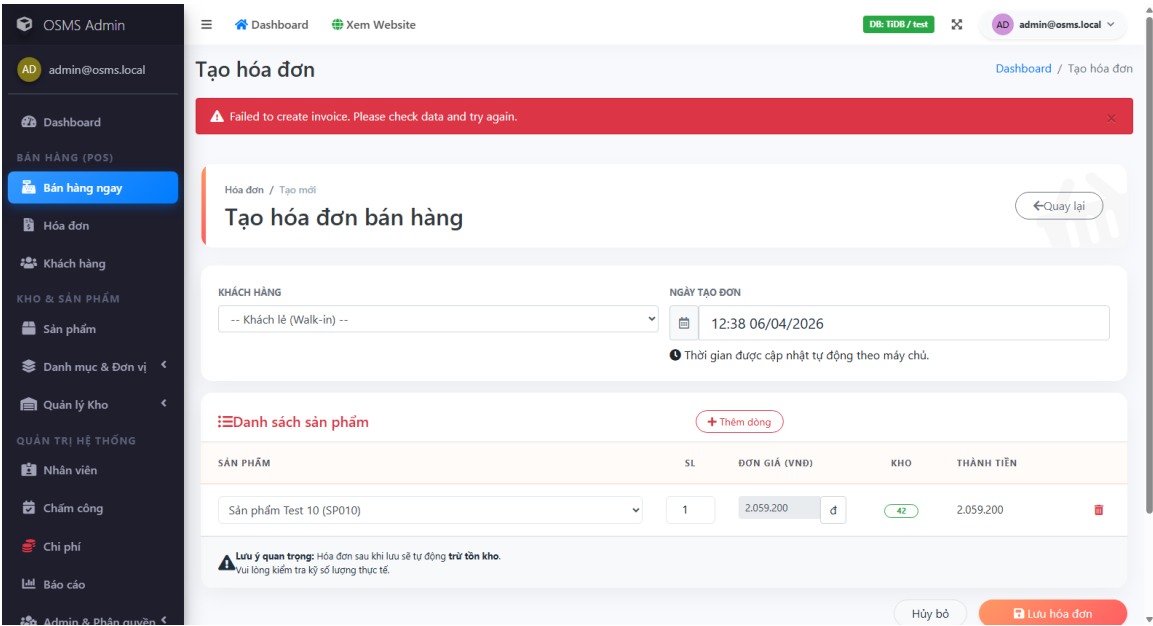


Figure E-4 - Invoice creation failure

Figure E-5. Permission-denial redirect evidence for sales user.

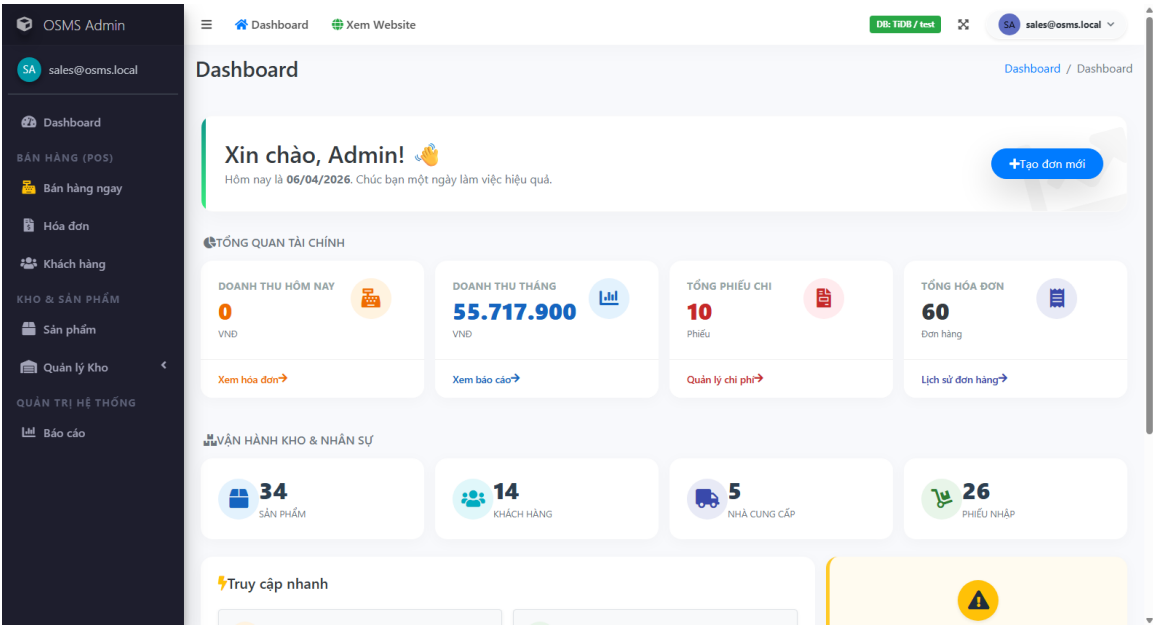


Figure E-5 - Permission denial

Figure E-6. Product import preview-count evidence.

OSMS Admin

warehouse@osms.local

Dashboard

KHO & SẢN PHẨM

Sản phẩm

Danh mục & Đơn vị

Quản lý Kho

Dashboard

Xem Website

DB: TEST / test

WA warehouse@osms.local

Xem trước import Excel

Dashboard / Xem trước import Excel

Xem trước dữ liệu import

Chỉ những dòng **hợp lệ** mới được import. Dòng lỗi sẽ hiển thị lý do.

← Upload lại

Danh sách

Tổng dòng

6

Hợp lệ

1

Lỗi

5

#	SKU	Tên	Danh mục	Đơn vị	Brand	Giá nhập	Giá bán	Tồn kho	Active	Trending	Trạng thái
2	QAIMP001	QA Import Product 01	30012 Laptop	30015 Cái	30039 Acer	100.000	120.000	10	✓	✗	Hợp lệ
3	QAIMP002		30012 Laptop	30015 Cái	30039 Acer	100.000	120.000	10	✓	✗	Lỗi • Name bắt buộc.
4	QAIMP003	QA Import Product Invalid Category	- UnknownCategory	30015 Cái	30039 Acer	100.000	120.000	10	✓	✗	Lỗi • CategoryName không tồn tại: UnknownCategory
5	QAIMP004	QA Import Product Negative Cost	30012 Laptop	30015 Cái	30039 Acer	-1.000	120.000	10	✓	✗	Lỗi • CostPrice không được âm.
6	QAIMPDUP	QA Import Product Duplicate 01	30012 Laptop	30015 Cái	30039 Acer	100.000	120.000	10	✓	✗	Lỗi • SKU bị trùng trong file Excel.
7	QAIMPDUP	QA Import Product Duplicate 02	30012 Laptop	30015 Cái	30039 Acer	100.000	120.000	10	✓	✗	Lỗi • SKU bị trùng trong file Excel.

Hủy

Import 1 dòng hợp lệ

Figure E-6 - Product import preview

IV. Defect Report And Metrics

4.1 Defect Log

At the current reporting date, the repository contains 4 confirmed product defects and 0 open observations pending manual confirmation. The two older automation observations remain closed after focused reruns proved that they were automation or expectation issues rather than product defects.

Current confirmed defect records:

- BUG-20260406-001
 - related to TC-UI-INV-001, TC-UI-INV-003, and TC-UI-INV-006
 - module: Invoices
 - severity / priority: High / High
 - current state: Open - Confirmed
 - GitHub Issue: #1
 - GitHub URL: <https://github.com/Alucard30Dec/Online-Sales-Management-System/issues/1>
 - evidence:
 - UI failure screenshot
 - focused rerun TRX
 - extracted server log excerpt showing `InvalidOperationException` in `InvoicesController.Create`
 - GitHub Issue screenshot with labels
- BUG-20260411-002
 - related to TC-UI-PUR-002 and TC-UI-PUR-003
 - module: Purchases
 - severity / priority: Medium / Medium
 - current state: Open - Confirmed

- evidence:
 - two UI failure screenshots
 - focused rerun TRX proving the validation banner is blank
- BUG-20260411-003
 - related to TC-UI-IMP-003
 - module: Product Import
 - severity / priority: High / High
 - current state: Open - Confirmed
 - evidence:
 - import-confirm failure screenshot
 - focused rerun TRX proving the valid preview still fails on confirm
- BUG-20260411-004
 - related to TC-UI-INV-005
 - module: Invoices
 - severity / priority: High / High
 - current state: Open - Confirmed
 - evidence:
 - invoice-cancel failure screenshot
 - focused rerun TRX proving cancellation does not complete

Record ID	Type	Related Test Case	Current Status	Evidence state
OBS-20260405-001	Observation	TC-UI-AUTH-003	Closed - Behavior Confirmed	rerun proved redirect-based denial is expected
OBS-20260405-002	Observation	TC-UI-IMP-002	Closed - Automation Fixed	rerun proved import preview works after locator fix
BUG-	Confirmed	TC-UI-INV-	Open -	UI screenshots,

20260406-001	Defect	001, TC-UI-INV-003, TC-UI-INV-006	Confirmed	TRX, server log, and GitHub Issue #1
BUG-20260411-002	Confirmed Defect	TC-UI-PUR-002, TC-UI-PUR-003	Open - Confirmed	UI screenshots and rerun TRX
BUG-20260411-003	Confirmed Defect	TC-UI-IMP-003	Open - Confirmed	UI screenshot and rerun TRX
BUG-20260411-004	Confirmed Defect	TC-UI-INV-005	Open - Confirmed	UI screenshot and rerun TRX

Figure B-1. GitHub Issue evidence for the confirmed invoice defect.

The screenshot shows a GitHub issue page for the repository 'Alucard30Dec / Online-Sales-Management-System'. The issue title is 'Invoice creation fails for valid walk-in invoice with in-stock item #1'. The issue is open and was created 5 days ago by user Alucard30Dec. The issue description includes the following sections:

- Environment:** Local test environment, <http://localhost:5068>, Google Chrome.
- Preconditions:** admin user logged in, /Admin/Invoices/Create reachable, active product SP010 with stock available.
- Steps to reproduce:**
 - Log in as admin@osms.local.
 - Open /Admin/Invoices/Create.
 - Keep customer as walk-in.
 - Add product SP010 with quantity 1.
 - Submit the invoice.
- Expected result:**
 - Invoice is created successfully.
 - Stock is reduced.
 - User is redirected to invoice details.
- Actual result:**
 - Create action stays on the Create page.
 - Toast shows failed to create invoice. Please check data and try again.
 - Server log shows InvalidOperationException in InvoicesController.Create.
- Attachments prepared in repository:**
 - [evidence/ui/automation/20260406_053902_TC-UI-INV-001-failure.png](#)
 - [results/automation-ui/invoice-rerun.trx](#)
 - [evidence/defects/BUG-20260406-001-invoice-create-log.txt](#)

The right sidebar shows the issue is assigned to no one, has no labels, projects, milestones, relationships, or development information. The bottom of the page shows a sign-up prompt for GitHub.

Figure B-1 - GitHub issue screenshot

4.2 Defect Management Workflow

The defect-management process was prepared using a GitHub-compatible workflow with explicit labels for:

- severity
- priority
- status
- module
- interface
- defect type

The intended workflow is:

1. execute or re-execute a related test case
2. classify the result as **Confirmed Defect**, **Observation**, or **Automation Script Failure**
3. open a GitHub Issue only after a defect is reproducible
4. attach screenshots, runner output, and expected-versus-actual details
5. retest before closure

Label group	Values used
Severity	severity:critical, severity:high, severity:medium, severity:low
Priority	priority:p1, priority:p2, priority:p3, priority:p4
Status	status:new, status:triaged, status:in-progress, status:ready-for-retest, status:closed, status:rejected
Module	module:auth, module:permissions, module:products, module:product-import, module:stock, module:purchases, module:invoices, module:reports, module:catalog-api, module:health-api
Interface / type	interface:ui, interface:api, interface:automation, type:defect, type:observation, type:automation-script

4.3 Test Summary Metrics

The current metrics show that the submission is structurally strong in planning, traceability, and execution completeness, while the main remaining weakness has shifted from coverage depth to unresolved defect count.

Key metrics:

- executed test cases: 63 / 63
- pass rate on executed cases: 88.89%
- fail rate on executed cases: 11.11%
- blocked rate on executed cases: 0%
- documented scenarios: 42
- mapped scenarios: 42
- scenario execution coverage: 100%

Interface-wise view:

- Admin UI
 - total: 41
 - executed: 41
 - pass: 34
 - fail: 7
- Public UI
 - total: 3
 - executed: 3
 - pass: 3
 - fail: 0
- API
 - total: 19
 - executed: 19

• pass: 19

Module-wise view now shows that every module has direct execution evidence. The public API surface is fully executed and stable in the current baseline. Invoices, Purchases, and Product Import still contain the current confirmed defects, while the remaining modules are currently stable under the executed cases.

Interface	Total	Executed	Pass	Fail	Not Run	Execution Progress %
Admin UI	41	41	34	7	0	100
API	19	19	19	0	0	100
Public UI	3	3	3	0	0	100

Module	Total	Executed	Pass	Fail	Not Run	Execution Progress %
Authentication	3	3	3	0	0	100
Permissions	3	3	3	0	0	100
Purchases	7	7	5	2	0	100
Invoices	6	6	2	4	0	100
Product Import	4	4	3	1	0	100
Stock	3	3	3	0	0	100
Reports	2	2	2	0	0	100
Public Catalog	3	3	3	0	0	100

Coverage status	Scenario count
Executed	42
Designed Only	0
Total	42

Figure M-1. Metrics summary exported from the workbook.

OSMS Test Metrics Summary

Execution baseline synchronized to Powered by GPT package on 2026-04-11

Total Cases 63	Executed 63	Pass 56	Fail 7	Not Run 0
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Execution quality highlights

Execution Progress: 100%
Scenario Execution Coverage: 100%
Confirmed Defects: 4

Status interpretation

Pass = expected behavior confirmed by runtime evidence.
Fail = runtime evidence showed a mismatch between expected and actual behavior.
Not Run = 0 in the current synchronized baseline.

Figure M-1 - Metrics summary

V. Conclusion And Future Work

5.1 Achievements And Compliance

This submission achieved the following:

- completed a source-based project audit rather than a generic theory-based plan
- defined a structured test plan, scenario set, UI/API test cases, and reusable test data
- implemented a real automation framework for UI and API testing
- captured real execution evidence for both UI and API
- prepared a professional defect-management package and metrics pack
- maintained clear traceability between scenarios, test cases, results, evidence, and observations

5.2 Challenges And Limitations

The main limitation of the current package is no longer execution depth. All 63 designed test cases now have real runtime evidence. The remaining limitation is unresolved defect count: the system still contains four confirmed defects across invoices, purchases, and product import. All four defects are now mirrored into live GitHub Issues, and the invoice-create defect remains the strongest case because it also has server-log proof. Because of this, the report still cannot responsibly claim that the full system is stable.

The package still lacks two completions that would improve its final polish:

- focused defect retests were executed again on 2026-04-11, and all four confirmed defects still reproduced without any status downgrade
- basic cross-browser evidence now exists through an Edge smoke rerun of TC-UI-AUTH-001

5.3 Future Enhancements

The highest-priority next actions are:

1. fix and retest the invoice-create defect tracked in GitHub Issue #1
2. fix and retest invoice cancellation, purchase validation rendering, and import confirm after fixes are available

3. refresh the defect log, final results workbook, and metrics pack after those post-fix retests
4. expand cross-browser evidence beyond the current Edge smoke proof if time permits

5.4 Final Conclusion

Based on the current execution evidence, the Online Sales Management System now has full case-level execution coverage across the designed UI and API test suite. The package is strong in test design, evidence traceability, and automation support, and it now avoids the earlier weakness of large Not Run sections. The automation evidence is also stronger because a basic Edge smoke rerun now confirms TC-UI-AUTH-001 outside the primary Chrome baseline. However, the system still contains four confirmed defects, all mirrored into live GitHub Issues, and the invoice-create defect remains the strongest defect because it combines UI proof with server-log proof. Focused retests on 2026-04-11 showed that those four defects still reproduce, so the most defensible conclusion is that the project is execution-complete and submission-ready, but still short of a maximum-confidence quality claim until those confirmed defects are fixed and pass post-fix retest.

References

1. MainReport/UEF - Final.pdf
2. MainReport/SupportingDocs/phase-0-project-audit.md
3. MainReport/SupportingDocs/test-plan.md
4. MainReport/SupportingDocs/test-scenarios.md
5. MainReport/SupportingDocs/phase-7-automation-design.md
6. MainReport/SupportingDocs/phase-9-execution-and-evidence.md
7. MainReport/SupportingDocs/phase-10-defect-log-and-bug-management.md
8. MainReport/SupportingDocs/phase-11-final-results-and-metrics.md
9. MainReport/SupportingDocs/phase-12-analysis-and-insights.md
10. Source repository: <https://github.com/Alucard30Dec/Online-Sales-Management-System>

Appendix

Appendix A. GitHub Submission Link

- <https://github.com/Alucard30Dec/Online-Sales-Management-System/tree/main/Report%20Test%20subject/SV00123-ATU-A01>

Appendix B. Key Attached Artifacts

- Main report: MainReport/Powered by GPT - Software Quality Verification - Final Report.docx
- Final PDF: MainReport/Powered by GPT - Software Quality Verification - Final Report.pdf
- Presentation: MainReport/Powered by GPT - Software Quality Verification - Presentation.pptx
- UI test cases: TestCases/UI/OSMS-UI-Test-Cases.xlsx
- API test cases: TestCases/API/OSMS-API-Test-Cases.xlsx
- Scenario file: TestCases/Scenarios/test-scenarios.md
- Automation scripts: TestScript-Data/Automation/
- Test data: TestScript-Data/TestData/
- Final results: TestResults/FinalResults/OSMS-Final-Test-Results.xlsx
- Metrics: TestResults/Metrics/OSMS-Test-Metrics.xlsx
- Defect log: TestResults/Defects/OSMS-Defect-Log.xlsx
- UI evidence: TestResults/Evidence/UI/automation/
- API evidence: TestResults/Evidence/API/newman-full-run.txt
- Video: Videos/OSMS-Automation-Demo.mp4

Appendix C. Remaining High-Value Execution Gaps

- Focused retests on 2026-04-11 confirmed that all four live defects still reproduce; true post-fix retest evidence is still required.

- Basic cross-browser evidence is now included through an Edge smoke rerun for TC-UI-AUTH-001.